

Capability	ED-MRDT (this work)	SMIS (2023)	SPCImage NG (B&H)	PyBroMo (2016)	FluoSim (2020)	SMeagol (2016)	MicroVIP (2021)
Brownian dynamics (membrane particles)	Yes 13,090 particles 2D periodic	No	No	Yes 3D solution free diffusion	Yes 2D membrane	Yes 3D compartments	No
Multi-species reaction network	Yes 5 types + Hill feedback	No	No	No	Binding only	Yes Gillespie	No
Fluorophore photophysics (CTMC)	Yes 4 states PSS kernel	Yes advanced (best in class)	No (analysis only)	Basic 2 states	Yes blinking + bleaching	Yes activation + bleaching	No
FRET from exact inter-particle distances	Yes Forster per pair from BD positions	No	No	Yes fixed E per population	No	No	No
Confocal FLIM (TCSPC histograms)	Yes 256 bins, 25 ns Gibson-Lanni PSF	No widefield only	No analysis only	No timestamps only	No	No widefield only	Yes multiple modalities
Shot noise + dark counts	Yes intrinsic Poisson + 100 Hz dark	Yes	N/A	Yes	Yes	Yes	Yes
BD to Photophysics integration	Yes positions feed Forster rates	No BD	No	No reactions	No FRET	No photophysics feedback	No
Photophysics to FLIM integration	Yes CTMC feeds TCSPC bins	No FLIM	No simulation	No FLIM	No FLIM	No FLIM	No
Full chain: BD to FRET to FLIM	Yes	No	No	No	No	No	No
Ground truth channel (molecules + image)	Yes positions, bonds, species, E per frame	Partial positions only	No	Yes trajectories only	Yes positions only	Yes positions + reactions	No
Platform	Python / Numba open source	MATLAB open source	Commercial closed	Python open source	C++ / Qt open source	MATLAB / C open source	MATLAB / VIP open source